

openapi: 3.1.0

info:

title: Cloud Health Office – Provider Access API

version: 4.0.0

description: |

FHIR R4 Provider Access API for CMS-0057-F compliance.

Enables healthcare providers to access patient clinical and claims data held by the health plan, with SMART on FHIR authentication, patient consent enforcement, PHI encryption, and HIPAA audit logging.

****CMS-0057-F requires**** payers to make clinical and claims data available to in-network providers with an active treatment relationship, beginning January

****Supported Resources:**** Patient, Claim, ExplanationOfBenefit, Encounter, Condition, Observation – all mapped from the plan's internal systems and X12 EDI transactions to FHIR R4 with US Core profiles.

****References:****

- CMS-0057-F Final Rule
- Da Vinci PDex Implementation Guide
- US Core IG v3.1.1+
- SMART on FHIR App Launch v2.0

contact:

name: Aureliaware, Inc.

url: <https://cloudhealthoffice.com>

license:

name: Apache 2.0

url: <https://www.apache.org/licenses/LICENSE-2.0>

servers:

- url: <https://api.cloudhealthoffice.com/fhir/r4>
description: Production
- url: <https://sandbox.cloudhealthoffice.com/fhir/r4>
description: Sandbox (synthetic data)
- url: <http://localhost:3000/fhir/r4>
description: Local development

security:

- smart_on_fhir: [user/*.read, openid, fhirUser]

tags:

- name: Search
description: Search across FHIR resource types for provider-attributed patient

- name: Read
description: Read individual FHIR resources by ID
- name: Consent
description: Patient consent verification for provider data access
- name: Bulk
description: Bulk data operations for provider rosters

paths:

— Multi-Resource Search —————

/({resourceType}):

get:

operationId: searchResources

summary: Search FHIR resources

description: |

Search for resources of a given type for provider-attributed patients.

Patient consent is validated automatically when `patient` parameter is pr

Supported resource types: Patient, Claim, ExplanationOfBenefit,

Encounter, Condition, Observation.

Results are returned as a FHIR Bundle (searchset) with pagination.

tags: [Search]

parameters:

- name: resourceType

in: path

required: true

schema:

type: string

enum: [Patient, Claim, ExplanationOfBenefit, Encounter, Condition, Ob

description: FHIR resource type to search

- name: patient

in: query

schema:

type: string

description: Patient reference filter (e.g., `Patient/pat-12345`)

example: "Patient/pat-12345"

- name: date

in: query

schema:

type: string

description: Date range (FHIR date parameter, e.g., `ge2025-01-01`)

- name: status

in: query

schema:

type: string

```

    description: Resource status filter
- name: category
  in: query
  schema:
    type: string
  description: Category filter (for Condition and Observation)
- name: _count
  in: query
  schema:
    type: integer
    default: 20
    maximum: 100
- name: _offset
  in: query
  schema:
    type: integer
    default: 0
responses:
  '200':
    description: Bundle of matching resources
    content:
      application/fhir+json:
        schema:
          $ref: '#/components/schemas/Bundle'
        examples:
          patients:
            summary: Patient search results
            value:
              resourceType: Bundle
              type: searchset
              total: 2
              entry:
                - resource:
                    resourceType: Patient
                    id: "pat-12345"
                    meta:
                      profile: ["http://hl7.org/fhir/us/core/StructureDefin
                      name: [{ family: "Smith", given: ["Jane"]}]]
                      gender: female
                      birthDate: "1985-03-15"
          conditions:
            summary: Condition search results
            value:
              resourceType: Bundle
              type: searchset
              total: 1
              entry:

```

```

    - resource:
      resourceType: Condition
      id: "cond-001"
      clinicalStatus:
        coding: [{ system: "http://terminology.hl7.org/CodeSy
code:
        coding: [{ system: "http://hl7.org/fhir/sid/icd-10-cm
subject:
        reference: "Patient/pat-12345"
      onsetDateTime: "2020-06-15"

'401':
  $ref: '#/components/responses/Unauthorized'
'403':
  $ref: '#/components/responses/ConsentDenied'

```

— Read Single Resource ---

/ {resourceType} / {resourceId} :

get:

operationId: readResource

summary: Read a single FHIR resource

description: |

Read a single resource by type and ID. Patient consent is validated
if the resource is patient-specific.

tags: [Read]

parameters:

- name: resourceType

in: path

required: true

schema:

type: string

enum: [Patient, Claim, ExplanationOfBenefit, Encounter, Condition, Ob

- name: resourceId

in: path

required: true

schema:

type: string

responses:

'200':

description: The requested resource

content:

application/fhir+json:

schema:

type: object

example:

resourceType: Observation

id: "obs-bp-001"

```

    status: final
    category:
      - coding:
          - system: "http://terminology.hl7.org/CodeSystem/observatio
            code: vital-signs
    code:
      coding:
        - system: "http://loinc.org"
          code: "85354-9"
          display: "Blood pressure panel"
    subject:
      reference: "Patient/pat-12345"
    effectiveDateTime: "2025-11-15T10:30:00Z"
    component:
      - code:
          coding: [{ system: "http://loinc.org", code: "8480-6", disp
            valueQuantity: { value: 128, unit: "mmHg", system: "http://un
      - code:
          coding: [{ system: "http://loinc.org", code: "8462-4", disp
            valueQuantity: { value: 82, unit: "mmHg", system: "http://uni
'401':
  $ref: '#/components/responses/Unauthorized'
'403':
  $ref: '#/components/responses/ConsentDenied'
'404':
  $ref: '#/components/responses/NotFound'

```

— Consent Check —————

/Consent/{patientId}:

get:

operationId: checkProviderConsent

summary: Check provider's consent to access patient data

description: |

Verify whether the authenticated provider has patient consent to access data. Consent is checked automatically on all resource endpoints; this endpoint allows explicit pre-flight verification.

Consent requires both an active treatment relationship and explicit patient authorization per the provider's NPI.

tags: [Consent]

parameters:

```

- name: patientId
  in: path
  required: true
  schema:
    type: string

```

```

responses:
  '200':
    description: Consent status
    content:
      application/fhir+json:
        schema:
          type: object
          properties:
            patientId:
              type: string
            providerId:
              type: string
            consentGranted:
              type: boolean
            authorizedResourceTypes:
              type: array
              items:
                type: string
            consentScope:
              type: string
              enum: [treatment, payment, operations]
          example:
            patientId: "pat-12345"
            providerId: "prov-npi-1234567890"
            consentGranted: true
            authorizedResourceTypes: ["Patient", "Claim", "ExplanationOfBenefit"]
            consentScope: "treatment"

```

— Bulk Member Match —————

```

/Patient/$member-match:
  post:
    operationId: bulkMemberMatch
    summary: Match a member across plan systems
    description: |
      FHIR $member-match operation. Accepts member demographics and
      coverage information and returns the matched Patient resource
      with the plan's internal identifiers.

      Used by providers to discover patient IDs before querying
      clinical data, and by payers for Payer-to-Payer data exchange.
    tags: [Bulk]
    requestBody:
      required: true
    content:
      application/fhir+json:
        schema:

```

```

    type: object
  properties:
    resourceType:
      type: string
      const: Parameters
    parameter:
      type: array
      items:
        type: object
  example:
    resourceType: Parameters
    parameter:
      - name: MemberPatient
        resource:
          resourceType: Patient
          name: [{ family: "Smith", given: ["Jane"] }]
          gender: female
          birthDate: "1985-03-15"
      - name: CoverageToMatch
        resource:
          resourceType: Coverage
          subscriberId: "MBR-2024-001"
          payor: [{ reference: "Organization/payer-001" }]
      - name: Consent
        resource:
          resourceType: Consent
          status: active
          scope:
            coding: [{ code: "patient-privacy" }]
responses:
  '200':
    description: Matched patient
    content:
      application/fhir+json:
        schema:
          type: object
        example:
          resourceType: Parameters
          parameter:
            - name: MemberIdentifier
              valueIdentifier:
                system: "https://cloudhealthoffice.com/member-id"
                value: "pat-12345"
  '422':
    description: No match found
    content:
      application/fhir+json:

```

```
    schema:
      $ref: '#/components/schemas/OperationOutcome'
```

— SMART Configuration —————

/metadata:

```
  get:
    operationId: getCapabilityStatement
    summary: FHIR capability statement
    tags: [Search]
    security: []
    responses:
      '200':
        description: CapabilityStatement
        content:
          application/fhir+json:
            schema:
              type: object
```

/.well-known/smart-configuration:

```
  get:
    operationId: getSmartConfig
    summary: SMART on FHIR configuration
    tags: [Search]
    security: []
    responses:
      '200':
        description: SMART configuration
        content:
          application/json:
            schema:
              type: object
            example:
              authorization_endpoint: "https://login.cloudhealthoffice.com/oauth2/token_endpoint: "https://login.cloudhealthoffice.com/oauth2/token
              scopes_supported: ["user/*.read", "openid", "fhirUser", "launch/p
              capabilities: ["launch-standalone", "client-confidential-symmetri
```

components:

schemas:

Bundle:

```
      type: object
      properties:
        resourceType:
          type: string
        const: Bundle
      type:
```



```

    type: string
    const: searchset
total:
  type: integer
entry:
  type: array
  items:
    type: object
    properties:
      fullUrl:
        type: string
      format: uri
      resource:
        type: object

OperationOutcome:
  type: object
  properties:
    resourceType:
      type: string
      const: OperationOutcome
    issue:
      type: array
      items:
        type: object
        properties:
          severity:
            type: string
            enum: [fatal, error, warning, information]
          code:
            type: string
          diagnostics:
            type: string

responses:
  Unauthorized:
    description: Invalid or expired SMART on FHIR token
    content:
      application/fhir+json:
        schema:
          $ref: '#/components/schemas/OperationOutcome'
        example:
          resourceType: OperationOutcome
          issue: [{ severity: error, code: login, diagnostics: "SMART on FHIR t

ConsentDenied:
  description: Patient has not granted consent for this provider

```

```
content:
  application/fhir+json:
    schema:
      $ref: '#/components/schemas/OperationOutcome'
    example:
      resourceType: OperationOutcome
      issue: [{ severity: error, code: forbidden, diagnostics: "Patient con
```

```
NotFound:
  description: Resource not found
  content:
    application/fhir+json:
      schema:
        $ref: '#/components/schemas/OperationOutcome'
```

```
securitySchemes:
  smart_on_fhir:
    type: oauth2
    description: |
      SMART on FHIR (EHR Launch or Standalone Launch).
      Providers authenticate via their EHR system or directly
      using their NPI-linked credentials.
    flows:
      authorizationCode:
        authorizationUrl: https://login.cloudhealthoffice.com/oauth2/authorize
        tokenUrl: https://login.cloudhealthoffice.com/oauth2/token
        scopes:
          user/*.read: Read all resources for attributed patients
          user/Patient.read: Read patient demographics
          user/Claim.read: Read claims data
          user/ExplanationOfBenefit.read: Read EOBs
          user/Encounter.read: Read encounters
          user/Condition.read: Read conditions/diagnoses
          user/Observation.read: Read observations/vitals
          openid: OpenID Connect authentication
          fhirUser: FHIR user identity claim
```